DATA COLLECTION SYSTEMS

CANADIAN LIDAR

PosPro

POSITIONING & MOTION POST-PROCESSING

User Guide

For PosPro 1.3.1 (QGIS 3.34 to 4.x, with DCS Core)

This guide is installed with the plugin and describes exactly this version.
Open it from the ? button on the PosPro panel, the project menu, or Help > Plugins.

Plugins > Canadian Lidar > Data Collection Systems > PosPro


DATA COLLECTION SYSTEMS
POSPRO · POST-PROCESSING
SAINT JOHN TEST DRIVE ▾
1 DATA
16 min 00 s
2 BASE
SJB1 (csrs)
3 INSTALLATION
antenna 0.50, -0.49, -0…
4 PROCESSING
processed 2026-11-02 08…
5 QUALITY
95.6% fixed
6 CALIBRATION
antenna lever arms
7 EXPORT
SBET, text, layers
QUALITY
96%
FIXED
Fixed 95.6%
INS only 4.4%
95.6%
FIXED AMBIGUITIES
4.4%
BRIDGED BY THE IMU ONLY
0.051 m
FWD/BWD SEPARATION (95%)
0.336 m
LARGEST 3D Σ
26 s
LONGEST GNSS GAP
6.4 s
PROCESSING TIME
COLOUR THE MAP BY
Solution type
Findings – click one to see it on the map and in the plots.
✔ 95.6% of the drive has fixed ambiguities.
⚠ Longest GNSS gap bridged by the IMU: 26 s (2 gaps over 2 s). (14:13:02 UTC)
✔ Forward/backward separation: 95% below 0.044 m horizontally, maximum 4.362 m.
⚠ Passes disagree by more than 10 cm for 30 s. (14:13:00 UTC)
⚠ Estimated 3D accuracy worse than 10 cm for 15 s. (14:13:10 UTC)
i Estimated IMU biases at the end: accelerometers 0.0828, 0.4111, 0.1997 mg, gyros -2.134, 1.952, 1.090 deg/h.
OPEN THE QUALITY PLOTS
Processing: done

What's new in 1.1

New in 1.3.0 – Trajectory Review (step 6, between Quality and Calibration). Also: a real-time mode on the Processing page (the Certus's own solution when there is no base station) and GPX export. PosPro now needs the DCS Core plugin and keeps its work in the job's Processing folder (see the DCS Processing Workflow Spec).

New in 1.1.1

- **Menu:** Plugins › Canadian Lidar › Data Collection Systems › PosPro.
- **Window button** (⤢, top right of the panel): maximise, float as a window, or dock on the left or right.
- **Readable everywhere:** table headers, switches and dropdown lists are drawn by PosPro itself, so Windows styles can no longer turn them white.
- **Installation** is laid out like the mapper's Settings › Hardware: vehicle (van, truck, car) with plan, profile and rear drawings; Positioning and motion (GPS1, GPS2); Survey sensor; Vehicle (rear axle, time offset). When the rover files are read, the offsets fill in by themselves from the mapper's hardware.json, or else from the Certus's own installation record. Under GPS1, PosPro shows what the Certus reports: green when it matches, amber when it doesn't.
- **Data coverage bars** on the Data and Base pages, one row each for IMU, GNSS 1, GNSS 2, real-time solution and every base. Mint means good, amber degraded, red missing. Hover over a bar for times.
- **Base-station finder:** a progress bar shows the file, MB and elapsed time, with a Cancel button. Files are saved in the project's work/base folder and the station is added as soon as the download finishes; a message says where.
- **Precise orbits** come from ESA first (most complete), then IGS and CODE.
- **Rear axle:** measure it once per vehicle (tape to 5 cm). Mapper 2.9.1 has a field for it under Settings › Hardware, and PosPro reads it from the capture.

Version 1.1 makes trajectories more accurate on long baselines and adds two calibration tools; the panel now looks like the Mobile Mapper dashboard.

Feature	What it gives you	Where
Both Certus antennas in tight coupling	Twice the carrier-phase data; heading stays observable when parked or driving straight; 33 % smaller drift in a 30 s tunnel	Installation, automatic

Feature	What it gives you	Where
Ionosphere and troposphere estimation	Fixed ambiguities on bases 10–30 km away (6 % fixed became 96 % in testing)	Processing, Atmosphere
Signal-strength weighting	Weak signals near trees and buildings count less	Processing
Lever-arm solver	Checks the antenna offsets you measured, to about 1 cm	Calibration
Mobile Mapper look	Same colours, fonts, logo and solution colours as the dashboard	Everywhere

PosPro deals only with position and motion. Lidar boresight calibration is done in the point-cloud software, which reads the SBET PosPro exports.

What PosPro does

PosPro turns a Certus capture and a base station file into a smoothed 200 Hz trajectory inside QGIS, exported as SBET + SMRMSG or text.

- **Reads directly:** Certus logs (`.anpp`) and mapper captures (`.clraw`, including unfinished `.partial` files). It uses packet 20 (time and real-time attitude), 28 (raw IMU), 60 (raw satellite observations) and 61 (GPS ephemeris).
- **Processes:** tightly coupled (default) or loosely coupled GNSS/INS, or GNSS only; differential with a base station (PPK) or precise point positioning; forward and backward passes combined.
- **Base stations:** your own RINEX files, NOAA CORS and IGS downloads; positions from built-in PPP or NRCan CSRS-PPP.
- **Outputs:** SBET/SMRMSG, text files with chosen columns (including NB Stereographic easting/northing), a GeoPackage track, map layers and quality plots.

Processing runs in a separate engine program, so a fault in processing cannot close QGIS. The forward and backward passes run at the same time on two cores.

How the pieces fit

The drive goes in once; the base, the installation offsets and the calibration feed the same filter, and the outputs leave from the Export step.

```
flowchart LR
  A[".clraw / .anpp\nfrom the mapper"] --> B["Read files\nIMU, GNSS antenna 1
+ 2"]
  C["Base station\nRINEX or CORS download"] --> D["Base position\nCSRS-PPP or
built-in PPP"]
  B --> E["Tightly coupled filter\nforward + backward"]
  D --> E
  F["Installation\nantenna lever arms"] --> E
  E --> G["Combined trajectory\n+ quality checks"]
  G --> H{"Offsets\nchecked?"}
  H -- "not yet" --> I["Calibration\nsolve lever arms, compare, apply"]
  I -- "new offsets" --> F
  H -- "yes" --> J["Export\nSBET, text, GeoPackage"]
  J --> K["Point-cloud\nprocessing"]
```

Install (Windows, QGIS 3.34 or 4.x)

1. In QGIS: **Plugins > Manage and Install Plugins > Install from ZIP**. Install `dcs_core-1.0.1.zip` first (shared by PosPro, AlignPro and CloudPro), then `pospro-1.3.1.zip`, then restart QGIS once.
2. PosPro is under **Plugins > Canadian Lidar > Data Collection Systems > PosPro**, and a teal button on the toolbar opens the panel.
3. **New project** (project box, top right): choose the job folder that holds the mapper captures and a name. PosPro creates **Processing/<name>/** there (its own work in 01_PosPro) and offers the captures it finds as rover files; the raw data is never changed. **Open project** reopens one; the last project opens by itself.
4. This guide is installed with the plugin: the **?** button at the top right of the panel, **Help > Plugins > PosPro user guide**, or the project box menu open the copy that matches the installed version.

Windows may ask once to allow `pospro-engine.exe`. Downloads (ephemeris, precise orbits, CORS) need internet and are cached in the QGIS profile folder.

The eight steps

Work down the steps on the left of the panel; each shows a green tick when it is ready, amber when something needs a look.

1. **Data** – Add the `.clraw` (or `.anpp`) files of one capture; they are read at once. The panel shows the recording time, IMU rate and gaps, GNSS epochs and signals for both antennas, and warnings (for example GLONASS off, or epochs not on whole seconds). Set a start and end time to process part of a capture. Leave “Download broadcast ephemeris” on; without internet add a navigation file from your base.
2. **Base** – Add a RINEX file, or download the nearest NOAA CORS or an IGS station for the capture's days. Pick the base with the star (**Use this base**): nearest, covering 100 % of the drive. Then give it an accurate position (next section).
3. **Installation** – Choose the vehicle and the motion sensor (Certus Evo). Enter the offsets from the Certus centre in its aligned frame: X forward, Y right, Z down, in metres: primary antenna (required), secondary antenna, output point (for example the Lidar origin) and rear axle. If the capture holds the mapper's hardware file, **Use offsets from the capture** fills all of them in. Leave **Use the secondary antenna in tight coupling** on.
4. **Processing** – Keep **Tightly coupled** and **Forward and backward, combined**. Choose constellations, elevation mask (12°) and ratio test (3.0). Keep Atmosphere on **Automatic** and **Weight by signal strength** on. Keep the wheeled-vehicle constraint and zero-velocity updates on for road work. Press **Process**; progress shows at the bottom and **Stop** ends it.
5. **Quality** – Read the summary and findings, colour the map, open the plots (section below).
6. **Calibration** – Optional, after processing: solve the antenna lever arms from the drive (section below).
7. **Export** – Choose the folder, the point (IMU centre, output point or antenna), the rate, and SBET + SMRMSG, a text file and/or a GeoPackage layer. A short export note records frame, epoch, time system and offsets.

Each step's pill on the left takes the mapper's sensor colours: green done, amber needs a look, red a problem, grey not yet. The project box at the top right turns red while a job runs.

INSTALLATION

Offsets are measured from the centre of the Certus in its own aligned frame: X forward, Y right (starboard), Z down, in metres – the same frame as the Certus alignment page and the mapper's Settings.

VEHICLE: Car / van

MOTION SENSOR: Advanced Navigation Certus Evo

	X FWD (M)	Y RIGHT (M)	Z DOWN (M)
PRIMARY ANTENNA	0.500	-0.490	-0.400
SECONDARY ANTENNA	0.000	0.000	0.000
OUTPUT POINT	0.250	0.000	-0.350
REAR AXLE	0.000	0.000	0.000

USE THE SECONDARY ANTENNA IN TIGHT COUPLING

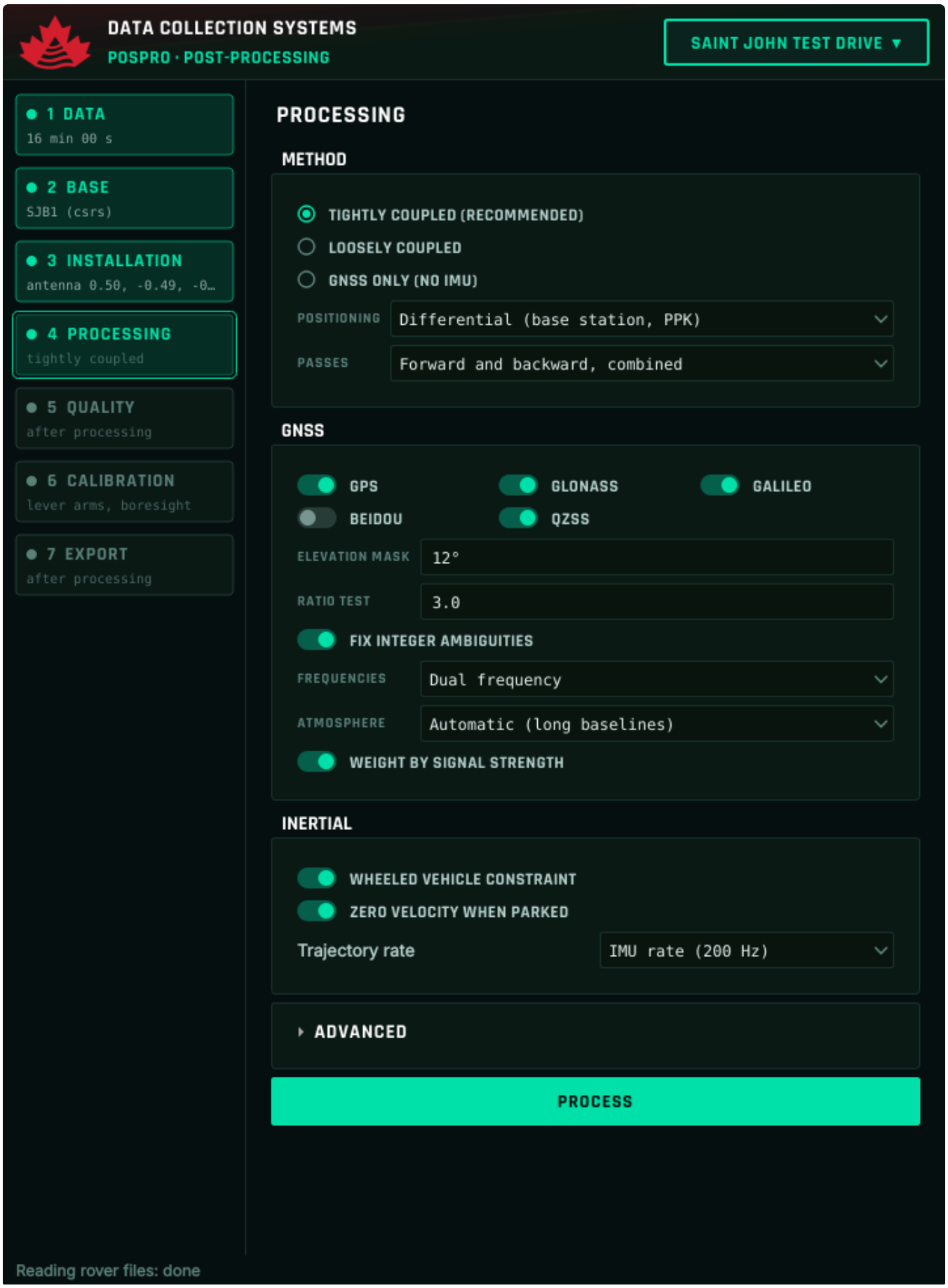
OUTPUT POINT NAME: LiDAR

IMU TIME OFFSET: 0.000 ms

USE OFFSETS FROM THE CAPTURE



The rear-axle point is used by the wheeled-vehicle constraint (no sideways slip). Leave it at 0 if unknown; it matters only in tight turns.


DATA COLLECTION SYSTEMS
POSPRO · POST-PROCESSING
SAINT JOHN TEST DRIVE ▾
1 DATA
16 min 00 s
2 BASE
SJB1 (csrs)
3 INSTALLATION
antenna 0.50, -0.49, -0…
4 PROCESSING
tightly coupled
5 QUALITY
after processing
6 CALIBRATION
lever arms, boresight
7 EXPORT
after processing
PROCESSING
METHOD
TIGHTLY COUPLED (RECOMMENDED)
LOOSELY COUPLED
GNSS ONLY (NO IMU)
POSITIONING
Differential (base station, PPK)
PASSES
Forward and backward, combined
GNSS
GPS
GLONASS
GALILEO
BEIDOU
QZSS
ELEVATION MASK
12°
RATIO TEST
3.0
FIX INTEGER AMBIGUITIES
FREQUENCIES
Dual frequency
ATMOSPHERE
Automatic (long baselines)
WEIGHT BY SIGNAL STRENGTH
INERTIAL
WHEELED VEHICLE CONSTRAINT
ZERO VELOCITY WHEN PARKED
Trajectory rate
IMU rate (200 Hz)
ADVANCED
PROCESS
Reading rover files: done

Base station positions

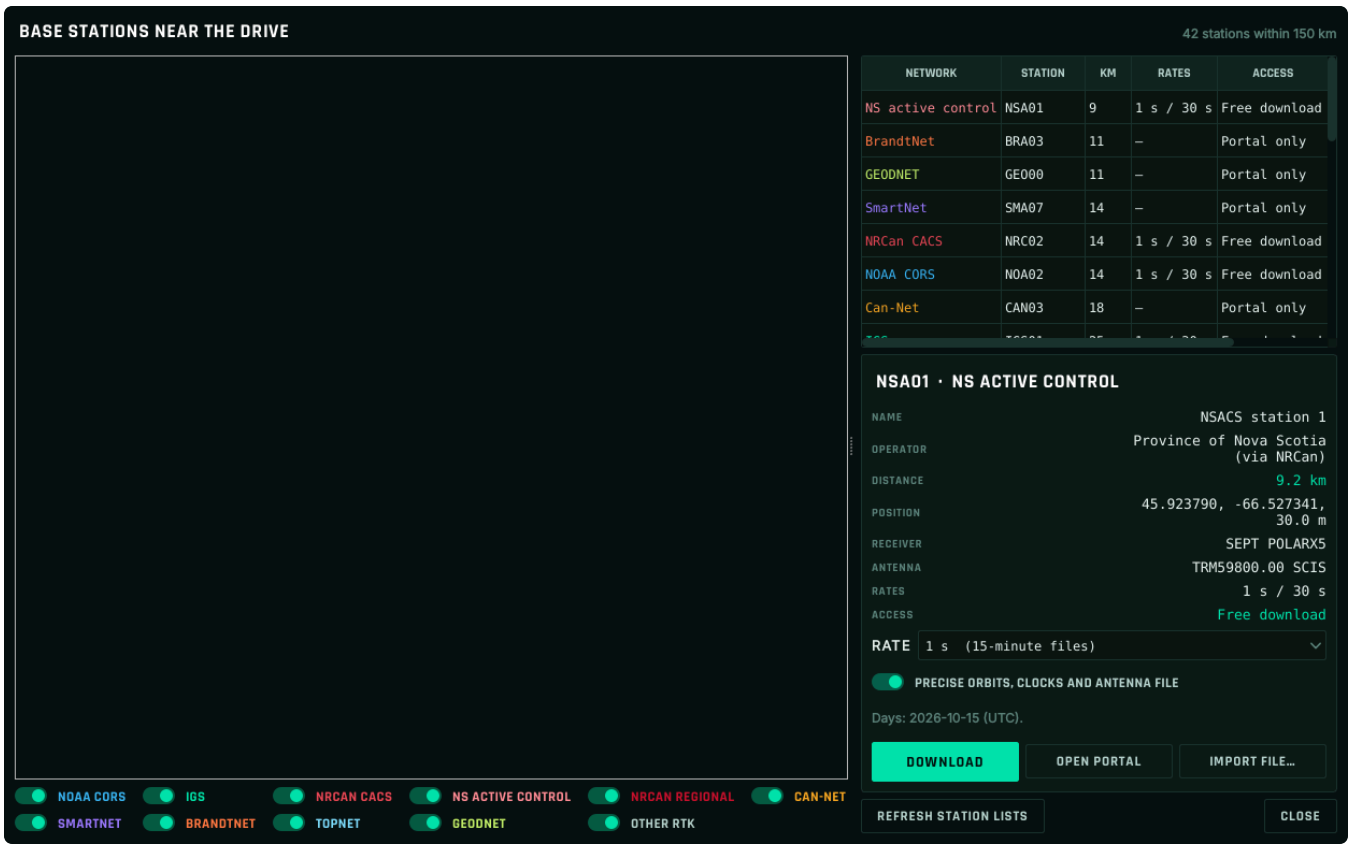
Every point of the trajectory inherits the base position's error and its frame, so give the base a precise position before processing. PosPro warns if the position still comes from the RINEX header.

Source	Use when	Result
NRCan CSRS-PPP	Internet and an NRCan account (your e-mail is the user name)	NAD83(CSRS) at the epoch you choose: New Brunswick 1997.0 (default), federal 2010.0, or the observation date
Built-in PPP	No CSRS account, or offline once orbits are cached	ITRF2020 at the observation date, converted to NAD83(CSRS) at the same date (EPSG:10415); no velocity applied
Entered / published	A surveyed or published coordinate	As you enter it, with its frame and epoch

- Coordinates are of the **marker**. The antenna height from the RINEX header and the antenna calibration (igs20.atx) are applied during processing.
- PPP needs at least 2 h of base data for a few centimetres and about 24 h for 1 cm. Final orbits appear about 2 weeks after the day, rapid ones after a day; PosPro uses whichever exists.
- For New Brunswick work in NAD83(CSRS) epoch 1997.0, use CSRS-PPP: the built-in PPP gives the observation date, which differs by the 1997–2026 crustal motion (centimetres).
- The frame and epoch of the base become those of the trajectory and of every export.
    - Rover PPP (no base, loosely coupled only) gives the trajectory in ITRF2020 at the observation date; PosPro labels the project frame that way.

Finding a base station

Download… on the Base page opens a map of every reference station within 150 km of the drive, so you pick the base by looking rather than by typing an ID.



Network	Symbol	Download from PosPro	Rates
NRCan CACS (FRDN, HLFX, UNBJ…)	red square	Yes, free (NRCan archive)	30 s daily, 1 s in 15-minute files
Nova Scotia active control	pink square	Yes, free (NRCan archive)	30 s, 1 s where recorded
IGS	mint star	Yes, free (BKG, IGN)	30 s daily, 1 s for high-rate stations
NOAA CORS	blue triangle	Yes, free (NOAA archive)	the station's own rate for 30 days, then 30 s
Can-Net, SmartNet, BrandtNet, TopNET, GEODNET, METACON	diamond, pentagon, hexagon…	No: **Open portal**, download there, then **Import file…**	per subscription

1. Read the rover files first; the map centres on the drive.

2. Click a station (or a row). The card shows its name, operator, distance (green within 20 km, amber 40, red beyond), receiver and antenna, rates and access. Commercial stations also show NRCan's check: compliant with NAD83(CSRS), unknown, or not compliant.
3. Pick the rate and leave **Precise orbits, clocks and antenna file** on, then **Download**. The files land in the project and the station appears in the base table, ready for CSRS-PPP or built-in PPP.

Station lists are cached for a week (**Refresh station lists** reloads them). The commercial stations come from NRCan's list of networks that agreed to tie their stations to NAD83(CSRS), so it covers Canada only.

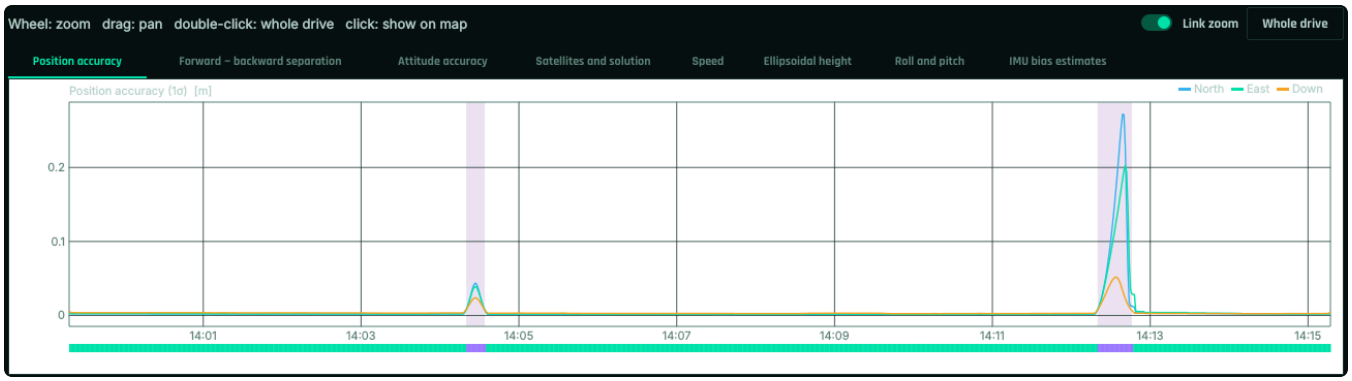
Reading the quality results

A good mobile-mapping trajectory is over 95 % fixed, has forward/backward separation under 5 cm (95 %), and no 3D accuracy worse than 10 cm outside GNSS gaps.

- **Donut and cards** – share of each solution type, time bridged by the IMU alone, separation, largest estimated 3D σ, longest GNSS gap, processing time. Green, amber and red follow the thresholds above.
- **Findings** – plain-language list; click one with a time to centre the map on it and zoom the plots.
- **Map** – layers in the "PosPro: <project>" group: real-time track (dashed), base stations with 20/40/60 km rings, the trajectory (1 point per second) coloured by solution type, estimated accuracy or separation.
- **Plots** (bottom panel) – position and attitude σ, forward − backward separation, satellites and ratio, speed, height, roll and pitch, IMU biases. A strip under each plot shows the solution type; GNSS gaps are shaded. Wheel zooms, drag pans, double-click shows the whole drive, a click marks the time on the map. Zoom is linked across tabs.

What you see	Likely cause	What to try
Fixed below 75 %	Base far away (over 40 km), poor sky, or a quarter-cycle L2 mismatch with the base receiver	A nearer base; L1 only; elevation mask 15°

What you see	Likely cause	What to try
Large separation at the start or end only	Vehicle parked: heading cannot be resolved without motion	Drive a few turns before and after the survey
Large separation in a long gap	IMU drift both ways; the combined result is weighted to the better pass	Expected; check the σ plot over the gap
Everything offset by the same amount	Base position from the RINEX header, or wrong frame	Set the base position (CSRS-PPP)
Processing stops at “could not align”	No packet 20 and the vehicle never drove faster than 4 m/s	Lower the alignment speed (Advanced)



Trajectory Review

After processing, step 6 lists every part of the trajectory that misses your limits, so you can look at each one and decide what to do with it before exporting.

1. **Set the limits.** Tick the checks you want: solution worse than fixed (or float, DGNSS), horizontal or vertical σ above a value, forward/backward separation above a value, IMU only for longer than a time, heading σ above a value. *Widen each by* adds a margin before and after each window, because receivers report a drop in quality a little late.
2. **Find anomalies.** The list shows what was found, when, how long, the worst value and the decision (open until you make one). The windows are drawn on the map in red. Double-click a row, or use **Zoom to**, to see it on the map and in the quality plots.
3. **Add your own. Rectangle**, **Polygon** or **Lasso** select part of the trajectory on the map (polygon: click the corners, right-click to finish). **Plot view** adds the time range the quality plots are zoomed to.
4. **Decide, for one row or several:**

- **Exclude GNSS** (tightly and loosely coupled) – the GNSS there is not used and the IMU carries the trajectory, as through a tunnel. Use it where multipath or a bad fix pulled the trajectory away. Press **Process again** to apply it; the map turns blue there.
- **Mark poor** – the trajectory is kept, but its σ is raised to the value you set, in the SMRMSG file and in what PosPro hands to AlignPro and CloudPro, so they can leave those points out or flag them. Amber on the map; applied at once.
- **Interpolate** (GNSS-only and real-time trajectories only) – a straight line between the good positions on both sides, horizontal and/or vertical. With the IMU, excluding the GNSS is always better, so this button only appears where there is no IMU solution. Violet on the map.
- **Accept** – looked at and fine as it is. Mint on the map.

5. **Undo**, **Reset to original** and **Edit log** are always there. Edits never touch the processed result itself: PosPro keeps an untouched copy and re-applies your decisions to it, so undo is instant and nothing is lost.

The step pill shows how many anomalies are still open, and amber while GNSS exclusions are waiting for **Process again**.

No base station? Choose **Real-time solution** on the Processing page. PosPro then uses the trajectory the Certus computed in the field (RTK if it had corrections), and Trajectory Review cleans it the same way, with Interpolate available.

Getting the most accuracy

Most of the accuracy is decided in the field; the software settings below then keep ambiguities fixed in conditions that used to defeat them.

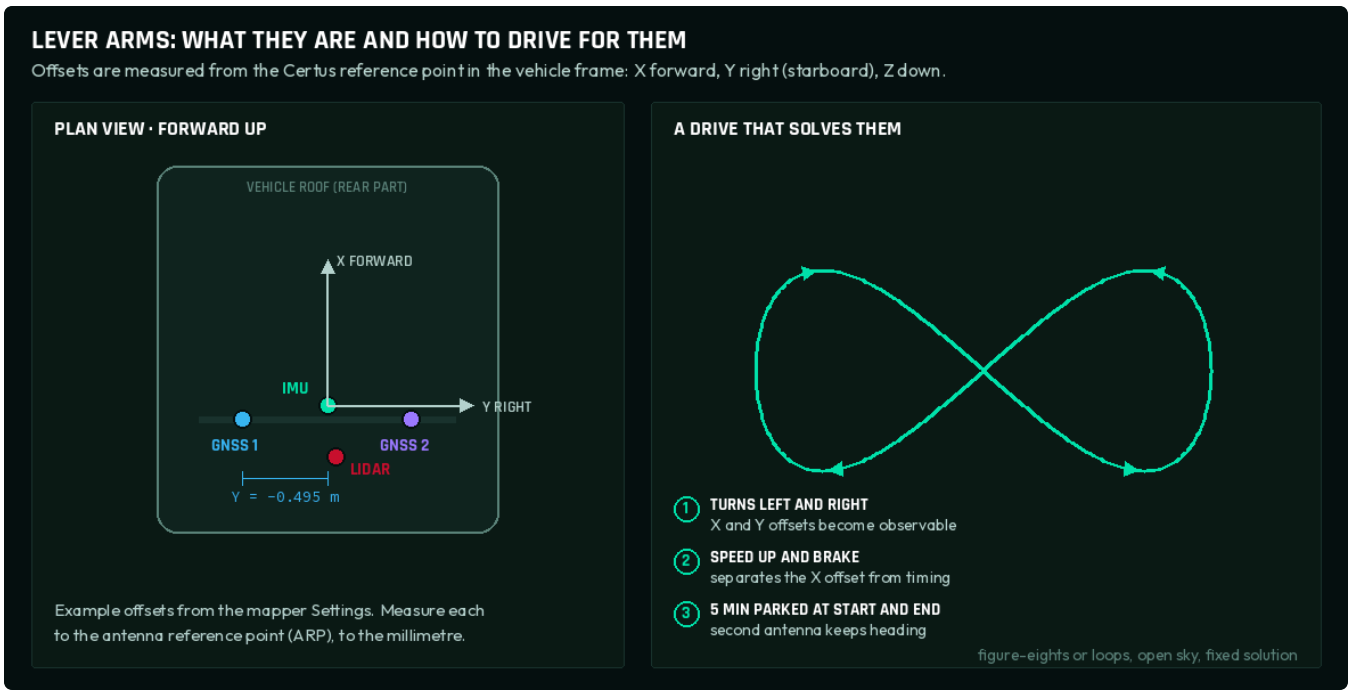
Setting or habit	Why it matters	Default
Both antennas recorded and used	Heading stays observable when parked or driving straight; more phase data in town	On when the secondary antenna's offset is entered

Setting or habit	Why it matters	Default
Atmosphere: Automatic	Beyond 10 km (or 150 m height difference) the ionosphere and troposphere differ between rover and base; estimating them keeps ambiguities fixable	Automatic
Weight by signal strength	Signals below 40 dB-Hz are often reflected or diffracted	On
GLONASS and BeiDou enabled on the Certus	More satellites in urban canyons; faster fixes after tunnels	Mapper “Recommended Certus outputs”
Base within 10–20 km, 1 s interval	Shorter baselines fix faster and hold through gaps	–
Offsets measured to the millimetre, then checked	A wrong offset moves every point and tilts the solution in turns	Calibration step
Turns and 5 min parked at start and end	Aligns the IMU and makes the biases and lever arms observable	–

Atmosphere estimation stays off on short baselines because there it only adds noise; in testing it made a 6 km drive slightly worse in height, which is why Automatic waits for 10 km.

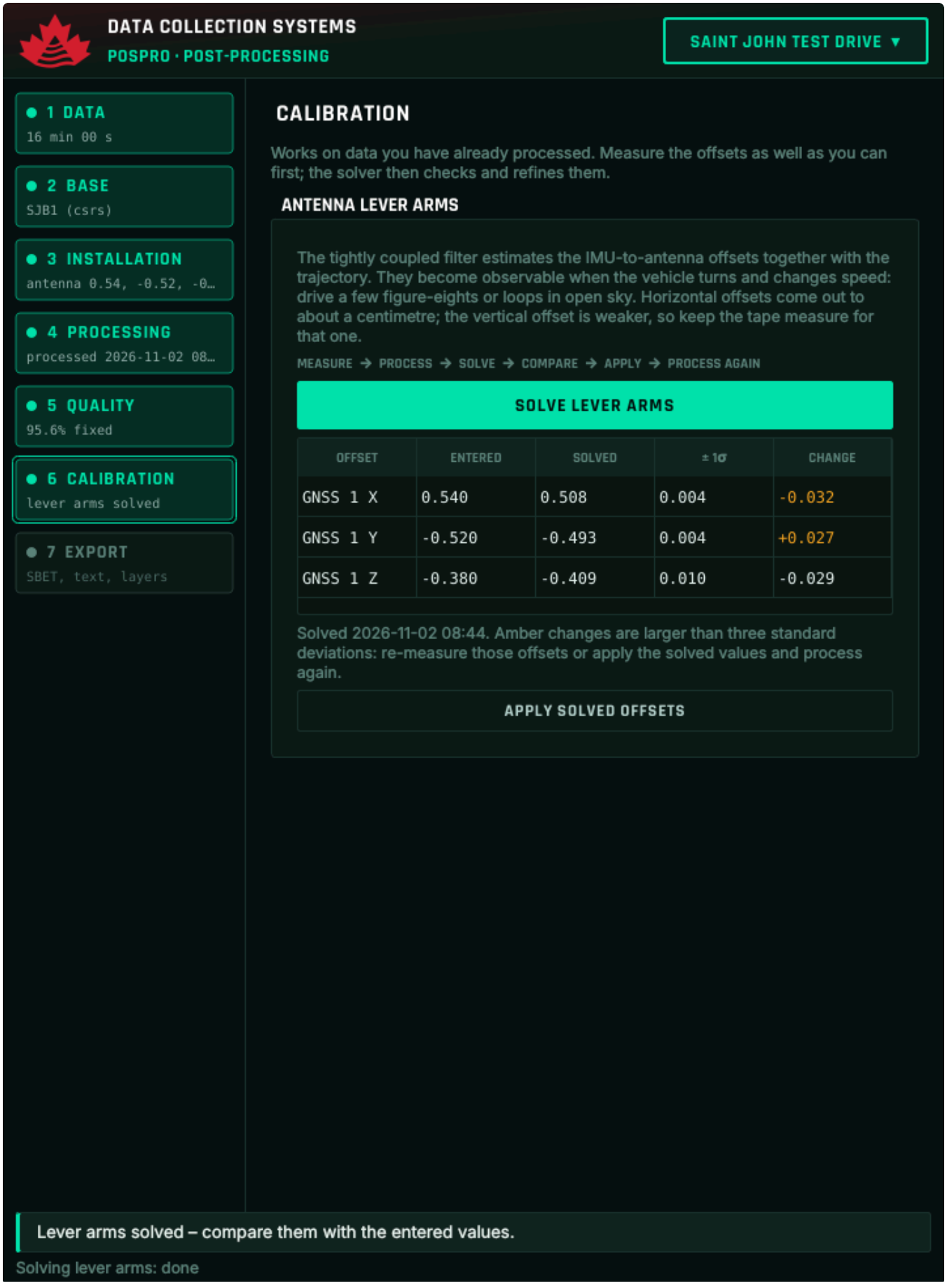
Calibration: antenna lever arms

The lever-arm solver estimates the IMU-to-antenna offsets from the drive itself and shows them beside the ones you entered, so a mis-measured antenna shows up before it spoils a survey.




```
flowchart LR
  A["Measure offsets\ntape, mm"] --> B["Enter them\nInstallation"]
  B --> C["Process\ntightly coupled"]
  C --> D["Solve lever arms\nCalibration"]
  D --> E{"Change larger\nthan 3 sigma?"}
  E -- "no" --> F["Keep the measured values"]
  E -- "yes" --> G["Re-measure that offset\nor apply the solved one"]
  G --> C
```

1. Process the drive with tight coupling and a fixed solution for most of it.
2. On **Calibration**, press **Solve lever arms**. Both passes run again with the offsets as unknowns (about as long as a normal processing run).
3. Read the table: entered, solved, ± 1σ and the change. Amber marks a change larger than three standard deviations and 2 cm.
4. Re-measure an amber offset if you can. Otherwise press **Apply solved offsets**, process again, and enter the same values in the Certus and the mapper Settings.


DATA COLLECTION SYSTEMS
POSPRO · POST-PROCESSING
SAINT JOHN TEST DRIVE ▾
1 DATA
16 min 00 s
2 BASE
SJB1 (csrs)
3 INSTALLATION
antenna 0.54, -0.52, -0…
4 PROCESSING
processed 2026-11-02 08…
5 QUALITY
95.6% fixed
6 CALIBRATION
lever arms solved
7 EXPORT
SBET, text, layers
CALIBRATION
Works on data you have already processed. Measure the offsets as well as you can first; the solver then checks and refines them.
ANTENNA LEVER ARMS
The tightly coupled filter estimates the IMU-to-antenna offsets together with the trajectory. They become observable when the vehicle turns and changes speed: drive a few figure-eights or loops in open sky. Horizontal offsets come out to about a centimetre; the vertical offset is weaker, so keep the tape measure for that one.
MEASURE → PROCESS → SOLVE → COMPARE → APPLY → PROCESS AGAIN
SOLVE LEVER ARMS
OFFSET
ENTERED
SOLVED
± 1σ
CHANGE
GNSS 1 X
0.540
0.508
0.004
-0.032
GNSS 1 Y
-0.520
-0.493
0.004
+0.027
GNSS 1 Z
-0.380
-0.409
0.010
-0.029
Solved 2026-11-02 08:44. Amber changes are larger than three standard deviations: re-measure those offsets or apply the solved values and process again.
APPLY SOLVED OFFSETS
Lever arms solved – compare them with the entered values.
Solving lever arms: done

Expect about 1 cm horizontally and 1–2 cm vertically from a drive with turns. A straight drive cannot separate the forward offset from the IMU's own errors, and the vertical offset is always the weakest, so measure it carefully.

Export and field practice

Export SBET + SMRMSG at the IMU centre (the Certus centre is the reference point of the mapper's survey configuration) and load `sbet_YYYYMMDD.out` in the point-cloud software.

File	Content
`sbet_YYYYMMDD.out`	17 doubles per record: GPS seconds of week, latitude, longitude (rad), ellipsoidal height, velocity N/E/D, roll, pitch, heading (rad), wander angle 0, body accelerations, body rates
`smrmsg_YYYYMMDD.out`	10 doubles: time, σ N/E/D (m), σ velocity, σ roll/pitch/heading (arc-minutes)
`<project>_YYYYMMDD.txt`	Columns you tick, any separator; UTC or GPS time; easting/northing in UTM, transverse Mercator or stereographic (EPSG:2953 by default)
`<project>_trajectory.gpkg`	The 1 Hz quality track for GIS

Heights are ellipsoidal; apply the geoid (e.g. HT2_2010v70) in the point-cloud software.

Field practice that improves the result

- Turn GLONASS on in the Certus (the mapper's "Set the recommended outputs"); raw observations then come every 125 ms on whole seconds.
- Log a base at 1 s, within 20 km, for the whole drive plus 10 minutes either side.
- Park 1–2 minutes, then drive a few turns at the start and the end; this resolves heading and IMU biases for both passes.
- Measure the antenna offset to the millimetre in the Certus aligned frame; an error there moves every point.

Verified accuracy and limits

On a simulated 16-minute drive with the Certus layout, tight coupling held 8 mm horizontal RMS with GNSS and 18 cm at worst in a 30 s tunnel. These are simulation figures; real data will be worse and must be checked against a reference.

Test (simulated, 16 min drive)	1.0 horizontal / vertical RMS with GNSS	1.1 horizontal / vertical RMS with GNSS	Fixed (1.0 → 1.1)	Worst in 30 s tunnel (1.0 → 1.1)
125 ms rover, 1 s base at 6 km	0.008 / 0.003 m	0.008 / 0.003 m	96 % → 96 %	0.18 → 0.18 m
Same, both antennas	–	0.008 / 0.003 m	96 %	0.18 → 0.12 m
72 ms rover, 30 s base at 15 km	0.022 / 0.043 m	0.009 / 0.022 m	93 % → 93 %	0.31 → 0.27 m
125 ms rover, 1 s base at 25 km, disturbed ionosphere (2 ppm)	0.95 / 1.32 m	0.005 / 0.012 m	6 % → 96 %	1.0 → 0.13 m
Loosely coupled, 6 km	0.020 / 0.048 m	unchanged	–	0.14 m

Calibration test (simulated)	Truth	Solved
Primary antenna lever arm, entered 3–4 cm wrong	0.500, −0.490, −0.400 m	0.508, −0.493, −0.409 m (± 0.004, 0.004, 0.010)
Both antennas, entered 3–7 cm wrong	primary as above; secondary 0.500, 0.490, −0.400 m	secondary 0.507, 0.487, −0.410 m

- 16 minutes of data processed in about 6 s; an 8-hour drive is estimated at a few minutes.
- The Windows engine gave results identical to the Linux build (run under Wine), including paths with accents and spaces.
- The panel was loaded and run in QGIS 3.34; the code is written for QGIS 4 (Qt6) but has not been run there.

Not yet tested: real Certus captures, live downloads (BKG, IGN, NOAA, IGS) and CSRS-PPP, real Windows and QGIS 4. The first real check: process one capture with a 1 s base and compare with Kinematica or Inertial Explorer on the same data.

Not included yet: network (multi-base) processing, odometer input. Lidar boresight and point clouds are left to the point-cloud software.